SDGs 11:
Sustainable Cities and Communities

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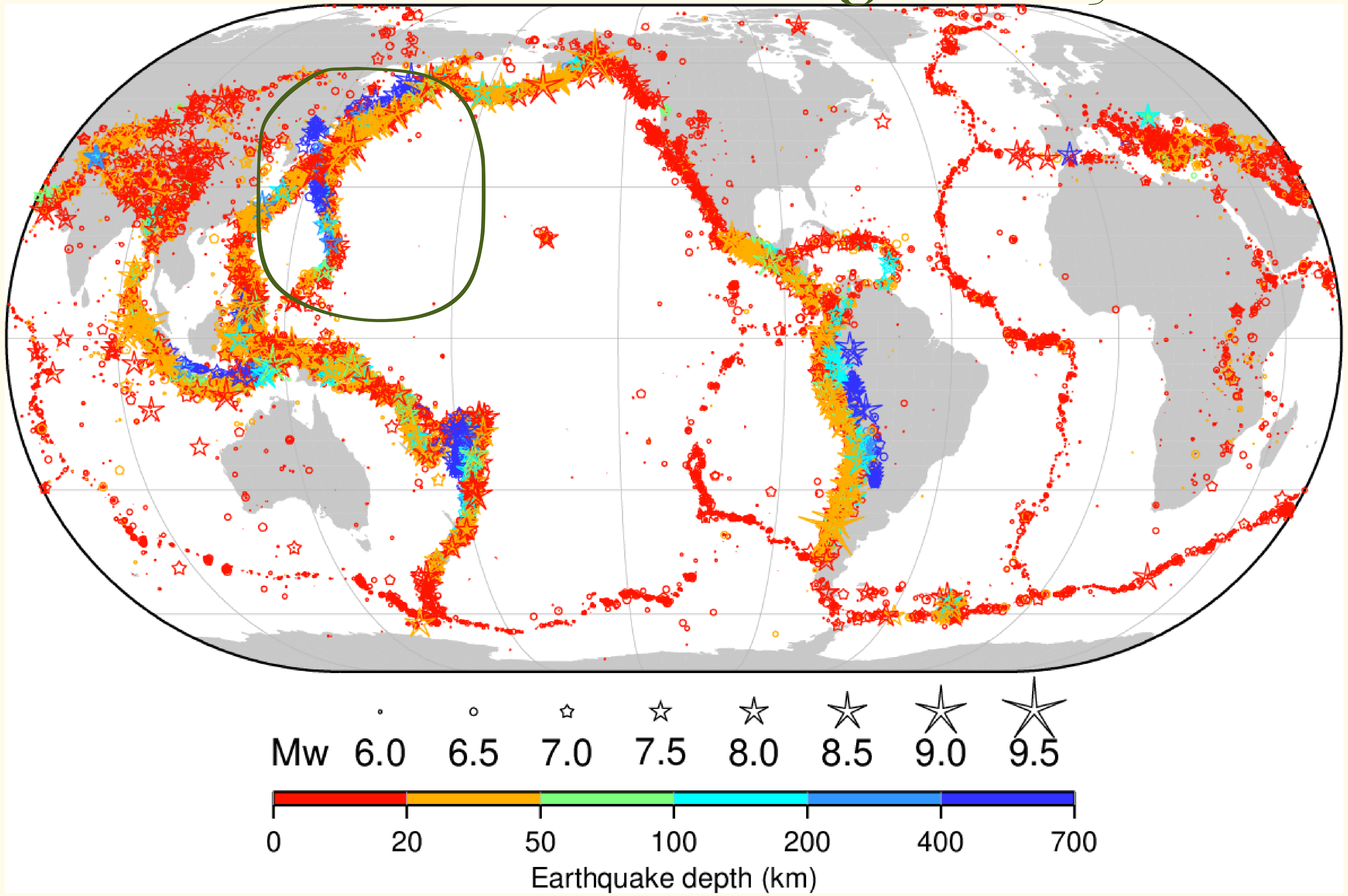
Sustainable Cities and Communities

- providing affordable housing
- improving sustainable transportation
- managing urbanization
- protecting cultural and natural heritage
- reducing disaster risks
- safeguarding the environment

Who is affected globally?

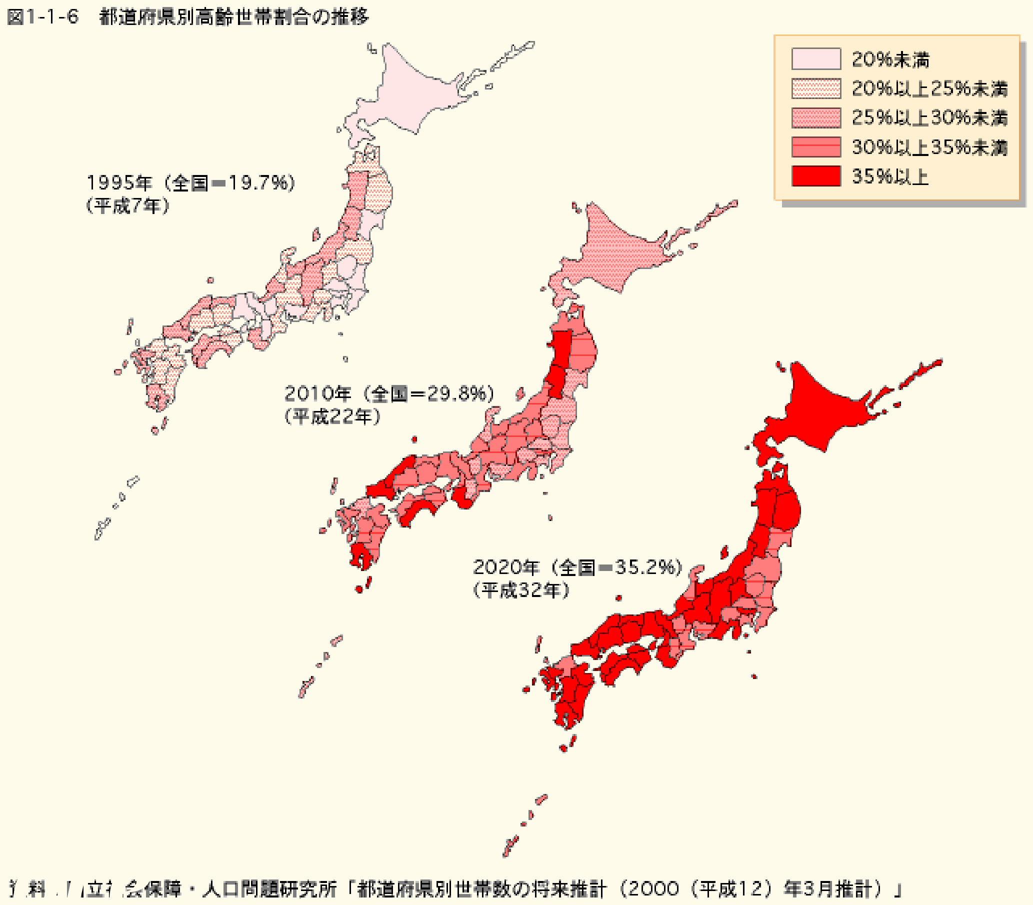


https://www.maxapress.com/app/figure_table/66433dc0fa6c58429b9216e3

Japan Earthquake Facts

- Four tectonic plates; Eurasian, Philippine, Pacific, North American intersect around Japan.
- 18.5% of M6 or higher earthquakes happen in Japan.

Different cities have different vulnerablity

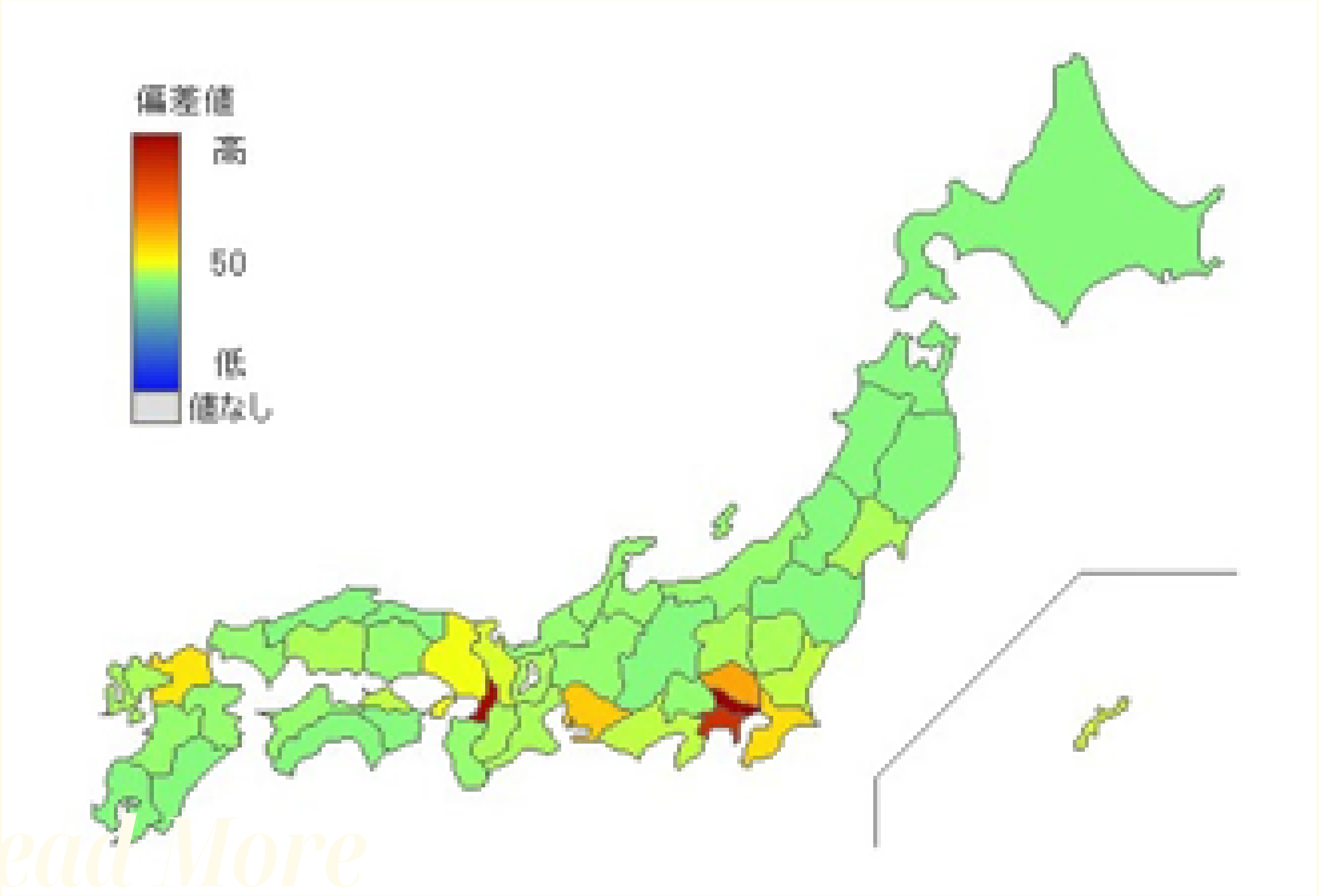


https://www.mhlw.go.jp/www1/wp/wp00_4/chapt-a1.html

Percentage of aged population

- Elderlies take longer to transport because of their illnesses, injuries, and incapabilities to drive
- They tend to have older house (especially in country side), which leads to easier collapsing due to deterioration. They were also built before earthquake resistent standards were set in 1981.

Different cities have different vulnerablity



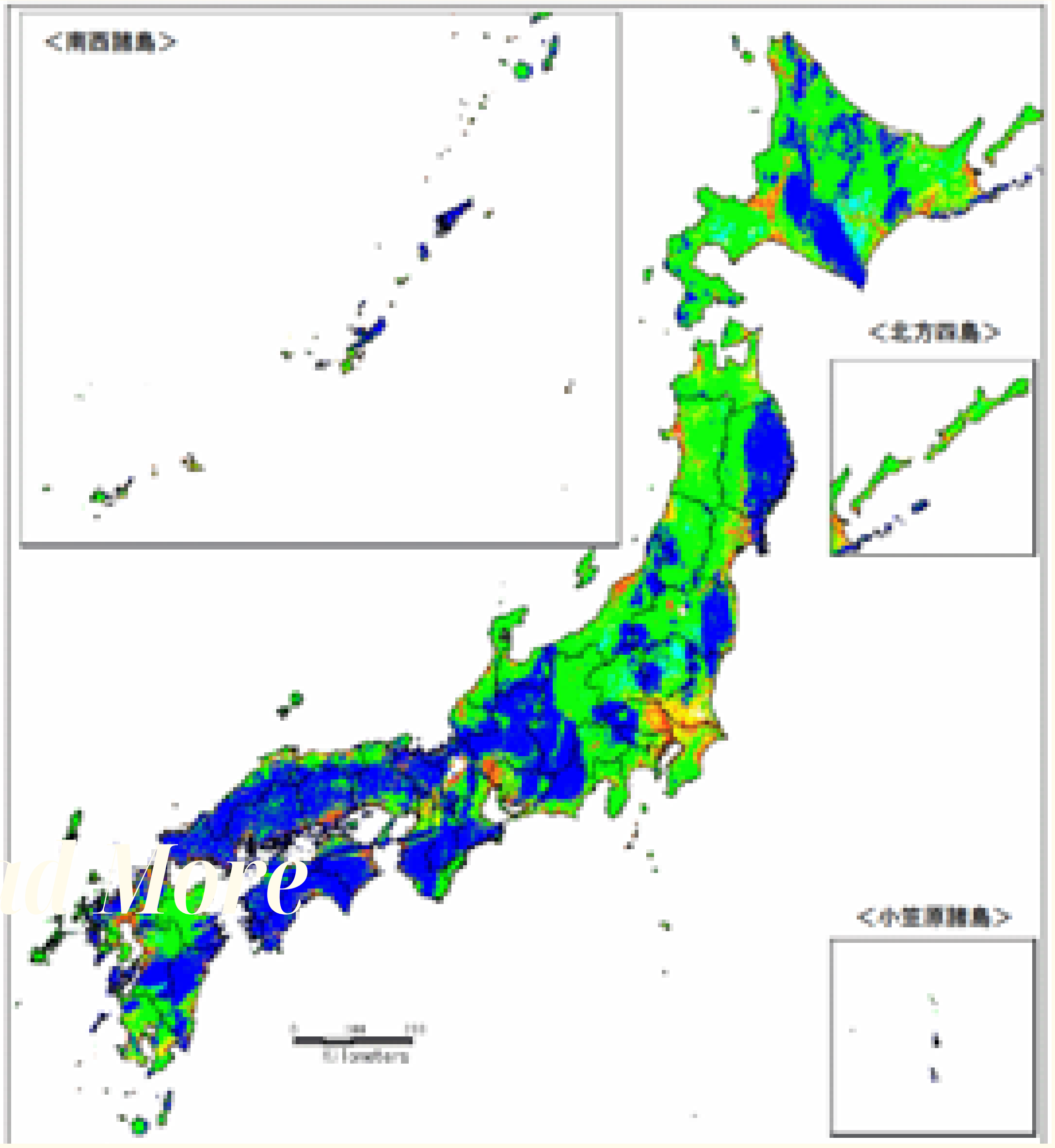
https://todo-ran.com/t/kiji/13400

Population density

Example: Greater Tokyo

- 30% of Japanese population live in 3.6% of the national land
- breakdown in small part of infrastructure affecting big population. (stranded commuters, hardships accessing resources)
- Packed wooden houses (especially in areas that has been developed for long time)

Different cities have different vulnerablity



https://www.jjjnet.com/jishin/jishin_yure_map.html

Firmness of ground

- Softer ground transmit more vibration

Earthquake effects

⚠️Real tsunami and earthquake pictures

Earthquake effects

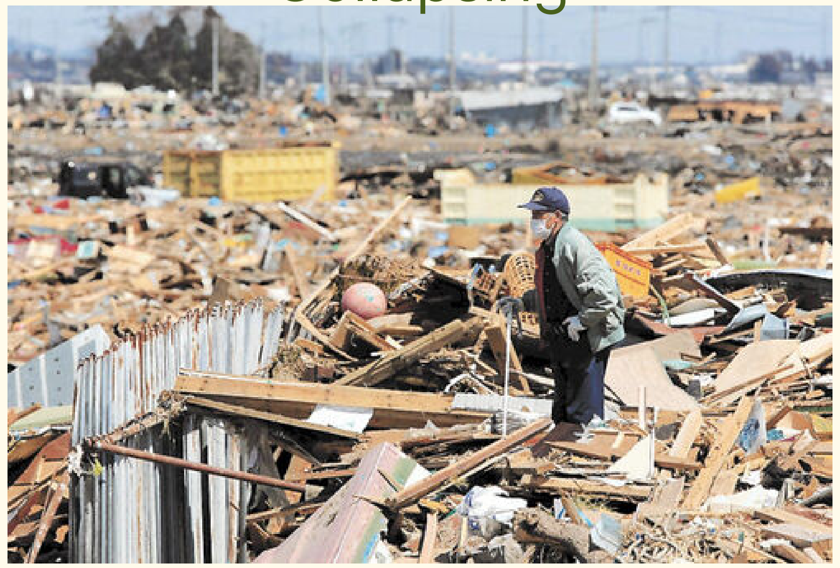
Tsunami

https://www.bousai.go.jp/kohou/kouhoubousai/h23/63/special_01.html

Nuclear pollution

https://www.jacom.or.jp/nousei/closeup/2022/220311-57469.php

Collapsing



https://kahoku.news/articles/20220314khn000019.html

Cracks

https://www.jiji.com/jc/d4?p=heq117-915826&d=d4_oo

Earthquake facts

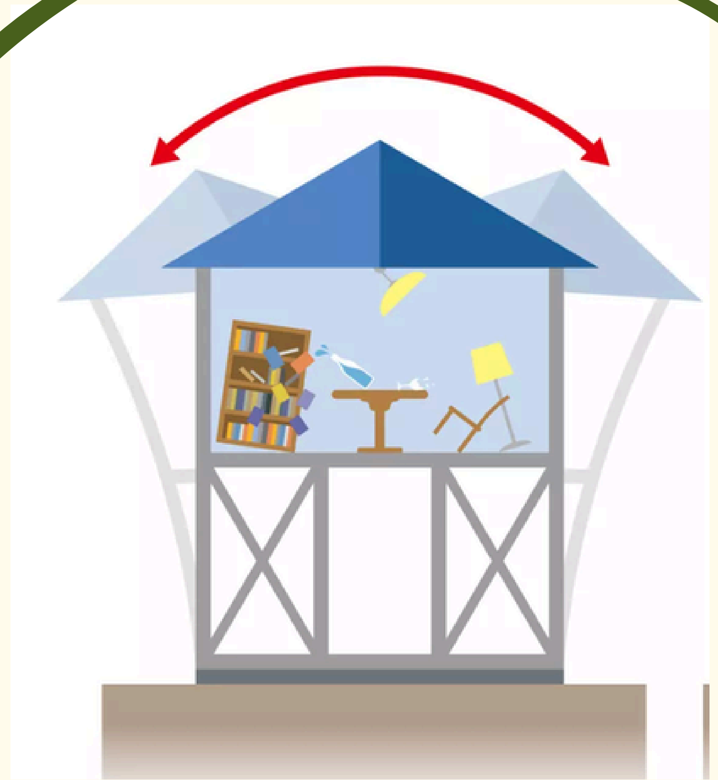
The Great East Japan Earthquake (2011)

- 15,900 deaths
- 2,525 Lost
- 470,000 Evacuated

The Great Kanto Earthquake (1923)

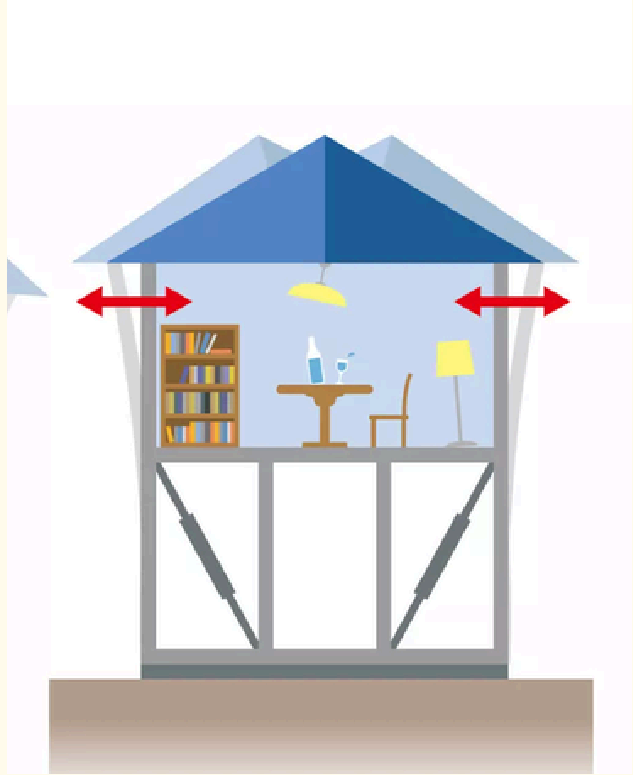
- 105,385 deaths
- 80% of death caused by fire

Solution #1: Earthquake proof buildings



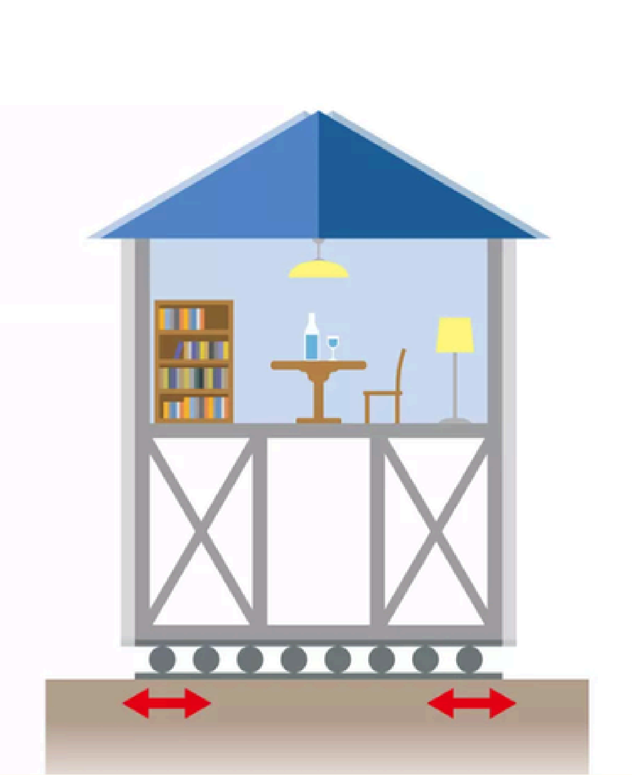
Earthquake resistant:

- stronger walls & structures
- low cost, low safety



Damping:

- absorbing movement with dampers
- medium cost, medium safety



Seismic isolation

- seperating the ground and the architecture
- high cost, high safety

<u>Most simple and adaptable in different places.</u>
<u>This can resist most of the earthquakes</u>

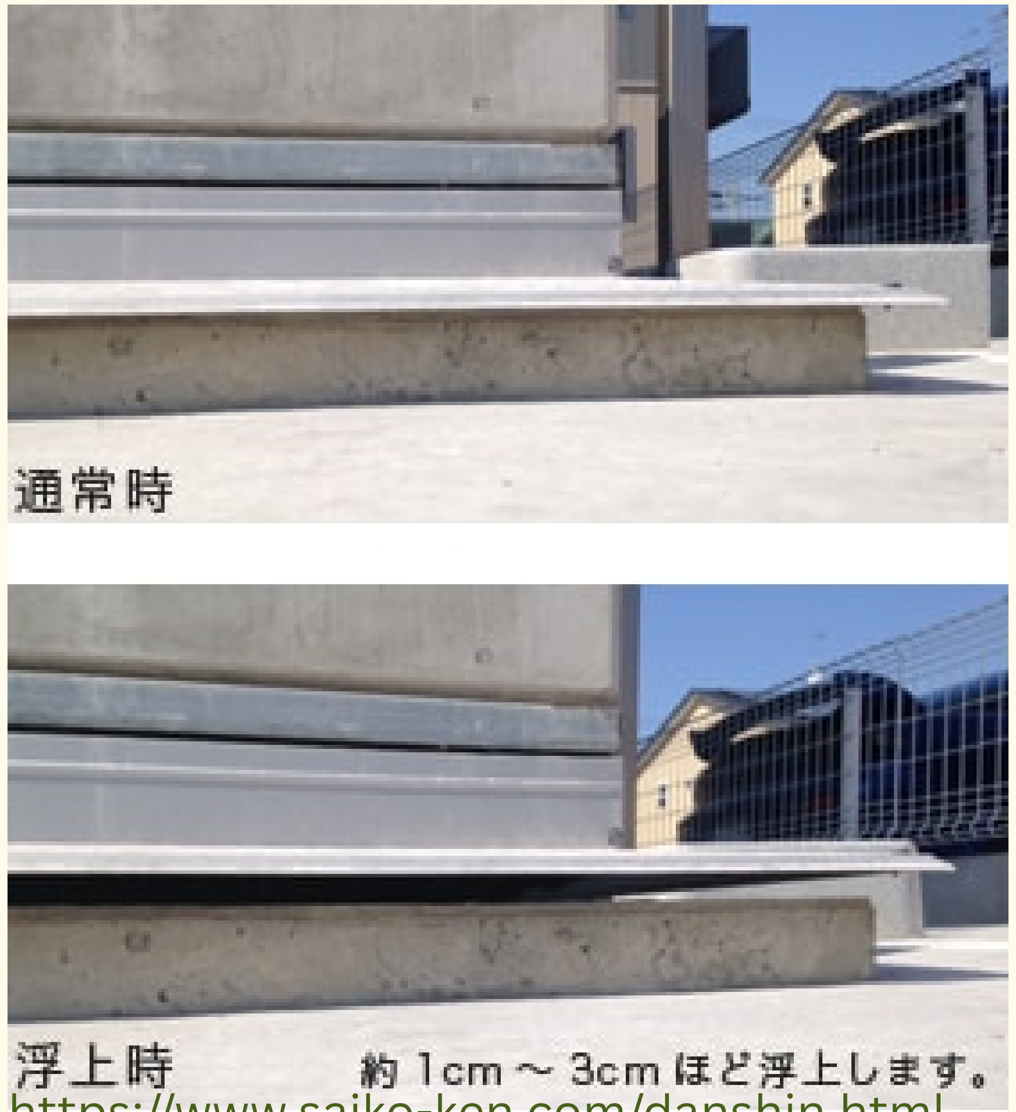
Solution #1: Earthquake proof buildings

By law, buildings in Japan are required to built to withstand Shindo 6-7 (around M7) since 1981

- Structure maps testing
- intermidiate testing
- Final testing

Solution #1: Earthquake proof building (new technology)

Air danshin: air makes the house float



https://www.saiko-ken.com/danshin.html

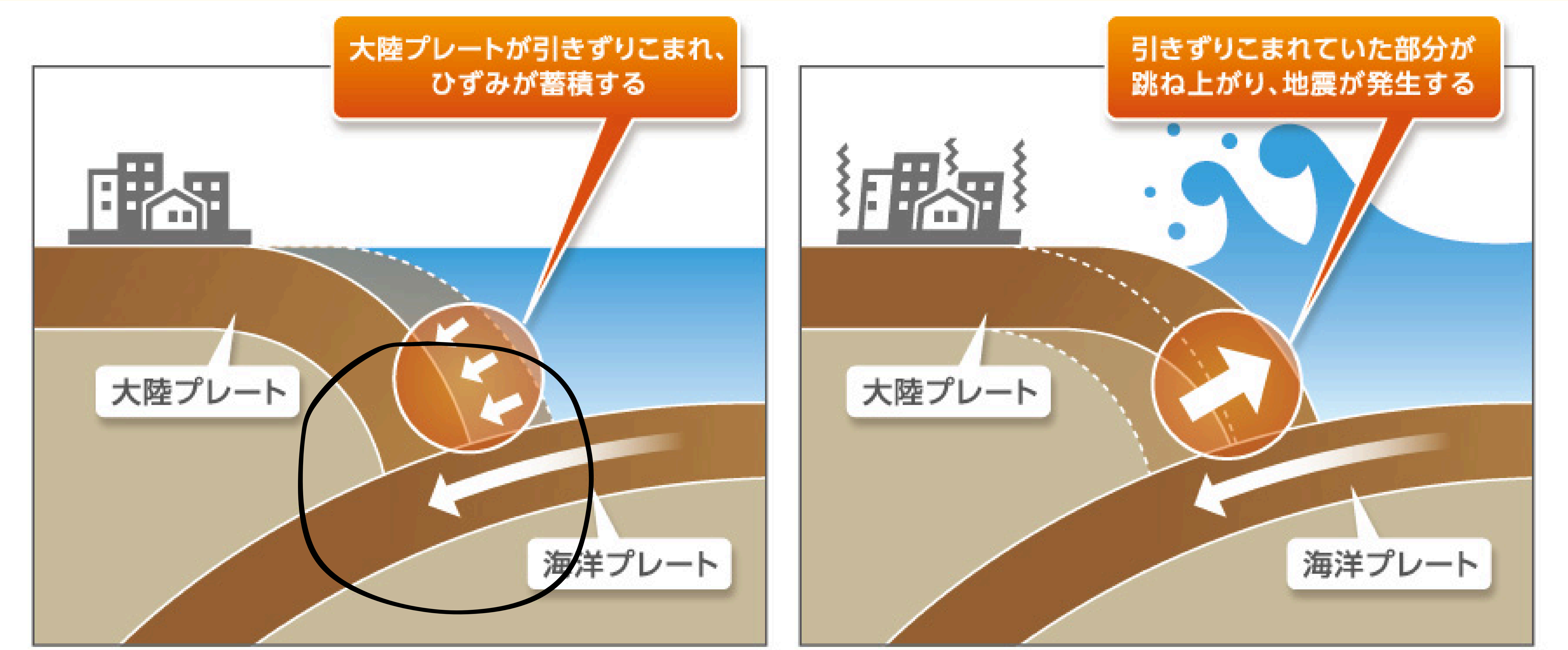
Active structural control: AI calculates and movezs counter weight to cancel out vibration from earthquakes

https://struc.kke.co.jp/works/512/

Limitation

Earthquakes cause other phenomenon like tsunami, which are almost impossible to resist

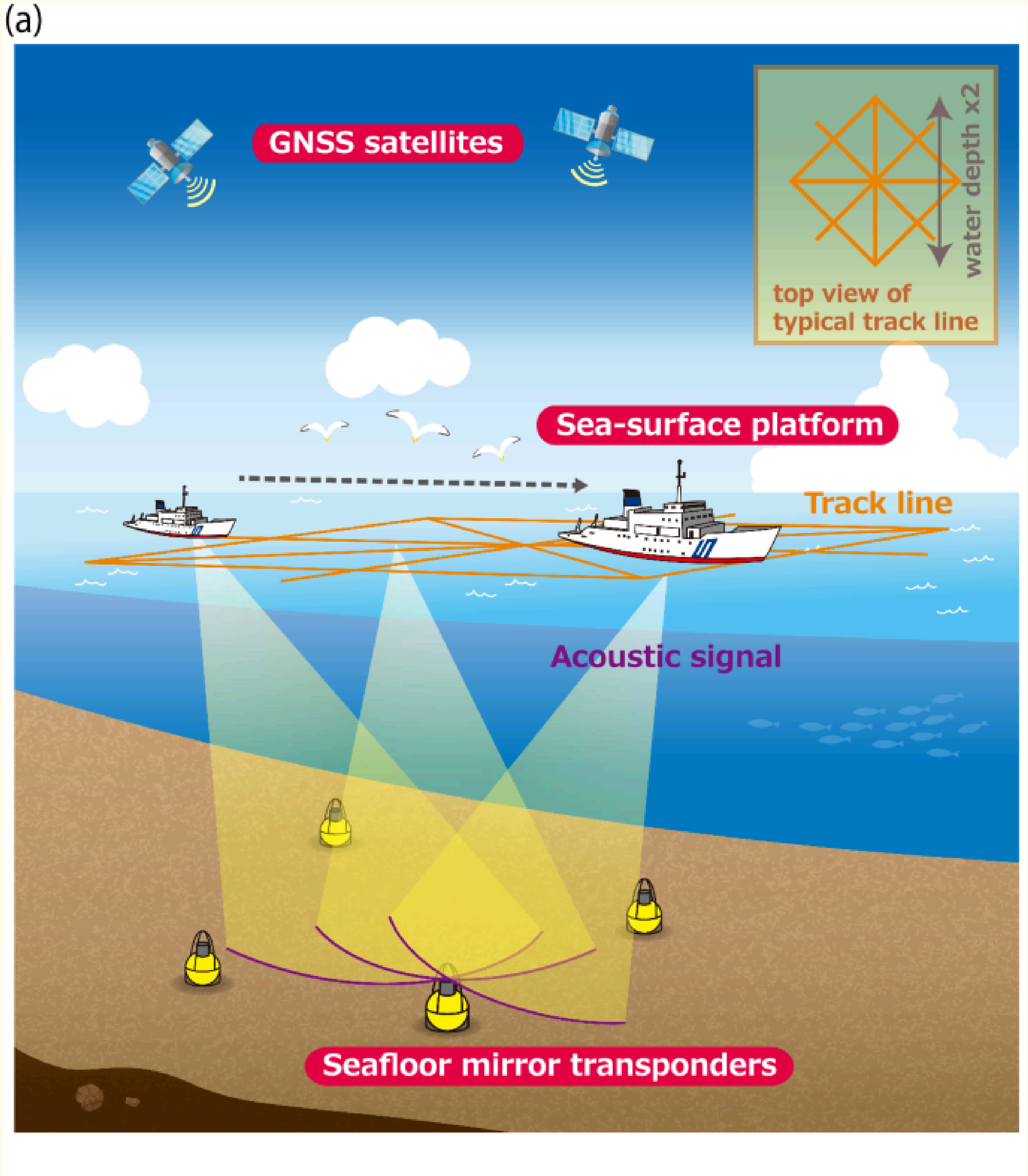
So what if we could predict earthquakes?



https://www.holdings.toppan.com/ja/bousai/shiru/03_16.html

<u>using lazer or sound to measure how much the plates are moving</u>

JCG GNSS-A



1. From Space: A ship (or a buoy) uses GPS satellites to find its exact position on the ocean surface.
2. To the Deep: The ship sends sound waves down to a sensor (transponder) sitting on the seafloor.
3. The Calculation: By measuring how long the sound takes to bounce back, scientists calculate the exact location of the seafloor sensor.

This system allows the Japan Coast Guard to:

- **Watch for Earthquakes:** It detects tiny movements (just a few centimeters) in the Earth's crust that land-based GPS cannot see.
- **Monitor "Silent" Movements:** It tracks "slow-slip" events where plates slide past each other without shaking, helping experts predict future giant earthquakes like the Nankai Trough.

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⭐JICE. "自然災害の多い国　日本." Www.jice.or.jp, www.jice.or.jp/knowledge/japan/commentary09.

⭐National Police Agency. "第1項　東日本大震災の被害状況及び主な警察活動." Npa.go.jp, 2021, www.npa.go.jp/hakusyo/r03/honbun/html/xf111000.html.

⭐Yokota, Yusuke, et al. "Experimental Verification of Seafloor Crustal Deformation Observations by UAV-Based GNSS-A." Scientific Reports, vol. 13, no. 1, 13 Mar. 2023, https://doi.org/10.1038/s41598-023-31214-6.

Thank You